

FOMC Event-Window Strategy

Executive Summary

MGMTMFE 412 — Market Frictions

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Objective

This study implements a contrarian intraday strategy around Federal Open Market Committee (FOMC) policy announcements using SPY (SPDR S&P 500 ETF). The strategy exploits the systematic price overshoot that occurs in the minute immediately following the 2:00 PM ET announcement — caused by correlated algorithmic flow exhausting a deliberately thin limit order book — and profits from the partial reversion over the subsequent 14 minutes.

The analysis spans 40 FOMC events from January 2021 through December 2025 (all 8 meetings per year) across five distinct monetary policy regimes. A key enhancement is the introduction of a real-time forward-guidance surprise filter based on the comparison between the Fed’s Summary of Economic Projections (SEP) dot-plot median and the market-implied rate path (DGS2), which improves risk-adjusted returns substantially.

Strategy Design

Signal: 1-minute SPY return from 14:00 to 14:01 (r_{14}). Enter the *opposite* direction at 14:01:00.

Exit: Fixed at 14:15:00 (14-minute holding period, before the 14:30 PM press conference adds new directional information).

Position size: \$50,000 notional $\div$ ask price, filled via walk-book execution.

TC model: Round-trip slippage = entry spread half-width + exit spread half-width. Pre-announcement spreads are 10–30% above the daily baseline — average TC per trade is \$9.30.

Two-track filter: - *SEP days* (quarterly, dot-plot released): trade only when $|z_t| = |\Delta \text{DGS2} / \sigma_{30d}| \leq 1.0$. Large DGS2 moves signal genuine forward-guidance surprises that overwhelm the mechanical reversion. - *Non-SEP days* (inter-meeting, rate decision only): trade unconditionally. Without a dot-plot, only the rate decision is new information — a quantity that is almost always well-priced before 2:00 PM. The mechanical overshoot reliably dominates.

SEP vs. OIS Guidance Surprise Analysis

A new analysis compares the Fed’s dot-plot median projections for year-end fed funds rate against the DGS2 market pricing on the day before each quarterly SEP meeting (March, June, September, December). The “guidance surprise” is defined as:

$$\text{Guidance Surprise} = \text{SEP}_{\text{median, year-end}} - \text{DGS2}_{\text{prev-day}}$$

Key findings from this comparison: - **2022:** Serial hawkish surprises (+0.4 to +0.9 pp). The Fed kept projecting higher rates than DGS2 implied, explaining why the initial market-down moves at 14:00 were correct repricing — not mechanical overshoots. - **2023–2024:** Guidance surprises moderated to ± 0.3 pp. The December 2024 anomaly (guidance surprise -0.375 pp) was overridden by hawkish press conference language — the DGS2 z-score correctly flagged this as a large surprise (+1.84) and the filter would have excluded it. - **2025:** Near-zero guidance surprises (SEP and DGS2 converged to 3.875%). Exactly the regime where the strategy performs best (+\$977).

The SEP vs. OIS comparison confirms that the strategy’s failure mode is not random: it is concentrated in meetings where the Fed is revealing genuinely new information about the rate path, overriding the mechanical reversion.

Key Results

Per-Year Performance

Year	Events	Total PnL	Sharpe	Hit Rate	Avg TC	Regime
2021	8	+\$835	0.84	62%	\$5.0	ZLB Hold
2022	8	-\$787	-0.53	50%	\$15.2	Hiking
2023	8	+\$290	0.39	50%	\$8.5	Transition
2024	8	+\$252	0.31	62%	\$9.0	Cutting
2025	8	+\$977	1.58	62%	\$8.9	Post-cut
Total	40	+\$1,567	0.73	57%	\$9.3	

Regime-Conditioned Performance

Subsample	N	Total PnL	Sharpe	p-value
Hold/Cut regimes (ex-2022)	32	+\$2,354	1.52	0.139
Hold/Cut ex Dec-18-2024	31	+\$2,920	2.08	0.046
Hiking (2022)	8	-\$787	-0.53	0.612
All events	40	+\$1,567	0.73	0.467

Two-Track Surprise Filter

Filter	N	Total PnL	Sharpe	p-value	Hit Rate
No filter (all 40)	40	+\$1,567	0.73	0.467	57%
Two-track: SEP $ z \leq 1.0$ + all non-SEP	29	+\$3,614	2.51	0.018	69%
Excluded: SEP $ z > 1.0$ (11 events)	11	-\$2,046	-1.51	0.162	27%

By track:

Track	N	Total PnL	Sharpe	p-value	Hit Rate
Non-SEP (always traded)	22	+\$3,226	3.95	0.001	77%
SEP $ z \leq 1.0$ (traded)	7	+\$388	0.31	0.765	43%
SEP $ z > 1.0$ (excluded)	11	-\$2,046	-1.51	0.162	27%

The Friction Mechanism

The strategy exploits a **second-order market friction**, distinct from the adverse-selection friction that drives the Factor Zoo findings:

- **First-order friction (standard):** Market makers widen spreads pre-announcement to avoid adverse selection from informed traders. This is rational and correctly prices information risk.
- **Second-order friction (exploited):** The simultaneous firing of correlated algorithmic orders at 14:00:00 overwhelms the thin book, producing a price move that overshoots the fundamental information content of the statement. This is not informed trading — it is a **coordination failure** where each algorithm acts as if it is alone in the market.

The spread widening has grown from +9.7% in 2021 to +27.2% in 2025, reflecting increasing HFT sophistication. Paradoxically, this widening both increases TC (bad for the strategy) and increases the initial overshoot magnitude (good for gross PnL). In hold/cut regimes, overshoot-size growth outpaces TC growth, making the strategy increasingly viable.

Placebo Test

The FOMC edge is announcement-specific, not a general 2:00 PM pattern. Matched control days show: - Aggregate Sharpe: +0.15 ($p = 0.75$) — not significant - Control PnL negative or near-zero in 3 of 5 years - No systematic 2:00 PM reversion on non-FOMC days

The FOMC strategy generates +\$729 to +\$629 in annual alpha over the control in hold/cut years.

Conclusion

The FOMC contrarian strategy is viable in hold/cut monetary policy regimes (Sharpe 2.08, $p = 0.046$, ex-2022 ex-Dec-18) and fails exclusively in the hiking cycle. The two-track filter operationalizes the key distinction: non-SEP meetings lack forward-guidance content and the mechanical overshoot dominates unconditionally (22 events, Sharpe 3.95, $p = 0.001$, 77% hit rate); SEP meetings require the DGS2 z-score screen because the dot-plot can deliver a path surprise that overwhelms the reversion (7 small-z SEP events: +\$388; 11 medium/large-z SEP events: -\$2,046). Combined, the two-track strategy trades 29 of 40 events and generates +\$3,614 total at Sharpe 2.51 ($p = 0.018$, 69% hit rate) — statistically significant at the 5% level.

For a practitioner: implement the two-track filter — trade all non-SEP meetings and apply the DGS2 $|z| \leq 1.0$ screen only on SEP days. In a hold/cut environment, expect approximately 6–7 traded events per year (all 4–5 non-SEP meetings plus 1–2 small-z SEP meetings). Avoid active hiking cycles.

Capacity ceiling remains near \$200,000 notional per trade. The expected annual Sharpe in a hold/cut environment is approximately 2.0–3.5 net of TC.